

RENEWABLE & INDUSTRIAL DECARBONISATION

Optimising Co-Located Renewable Power
and Green Fuel Hubs for Export and Mining
in the case of Chile

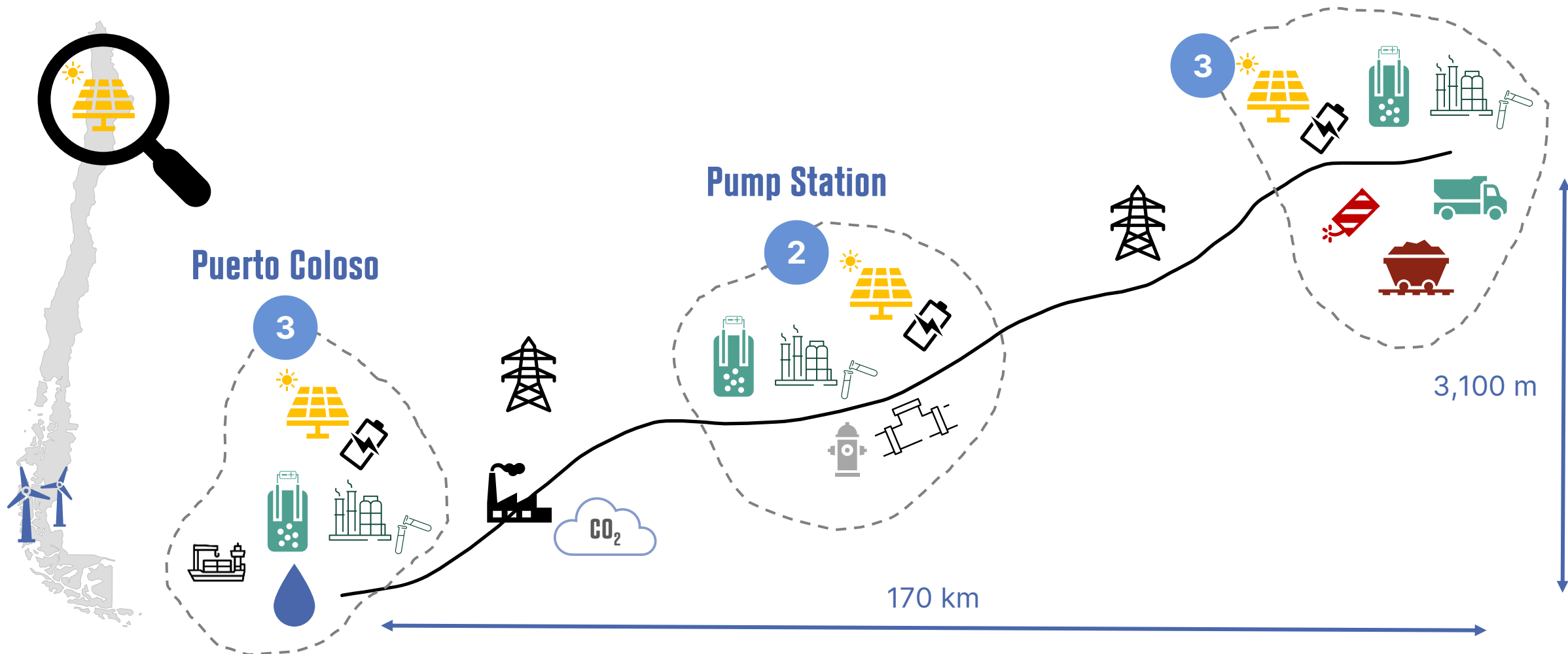


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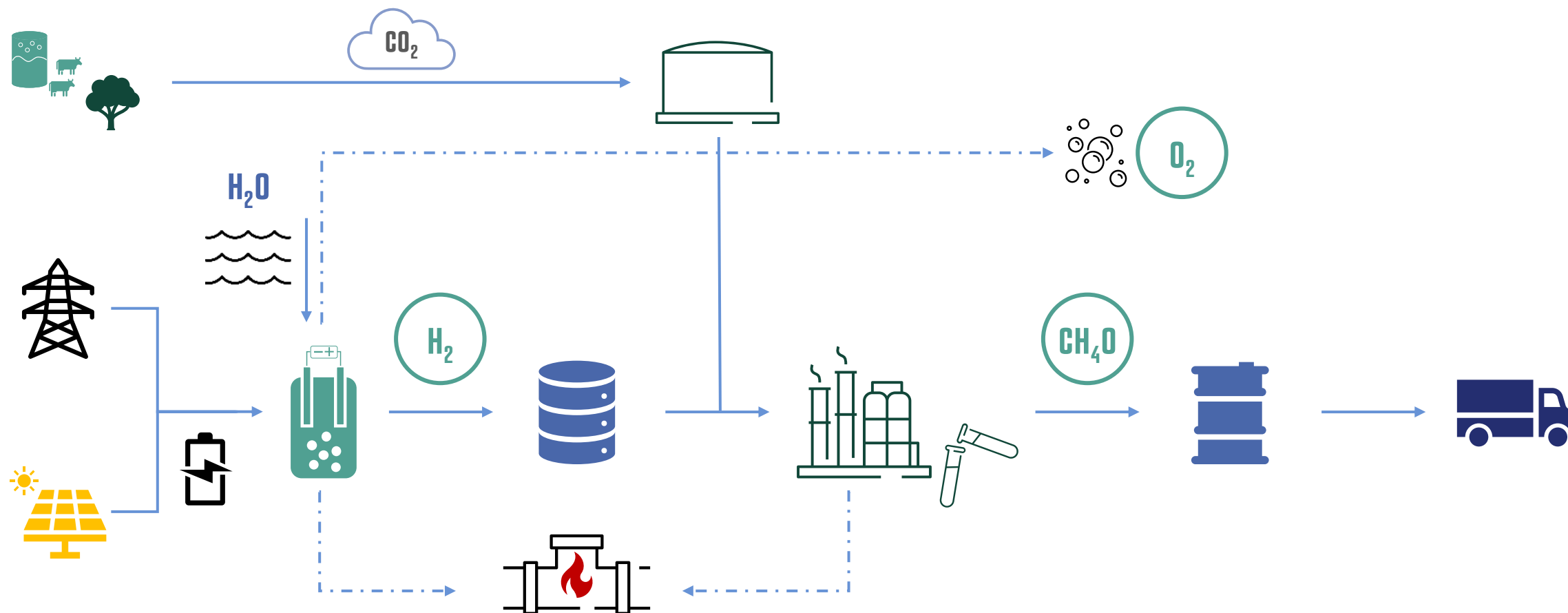
3 locations for ammonia and methanol production

Production at the mine, the export port or in between



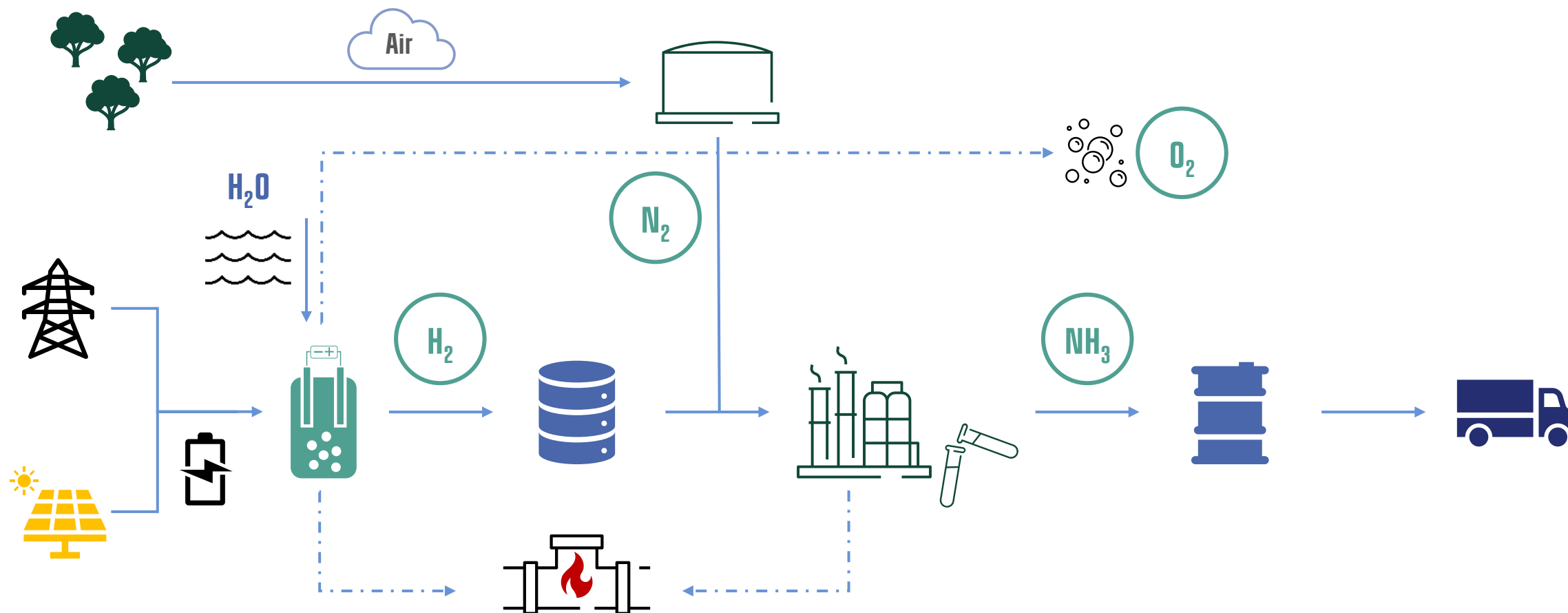
CSEI's Energy Hub Model

Methanol Production



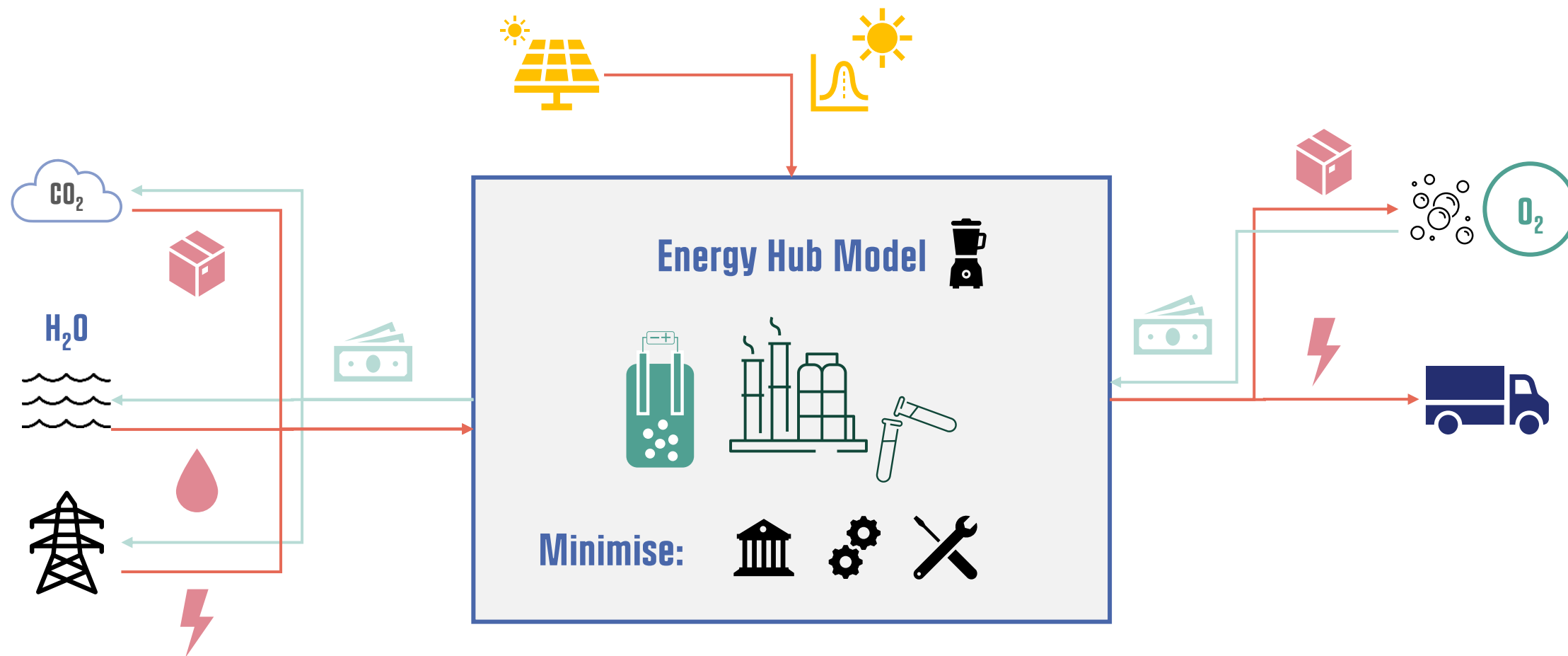
CSEI's Energy Hub Model

Ammonia Production



Deterministic Cost-minimising optimisation

Mixed-Integer Linear Program considering efficiency curves for electrolysis



Ammonia & Methanol PtX-Production in the Atacama Desert

Input Data and Assumptions for Fuel Production

▪ Scenarios

- **Methanol production:** domestic demand for fuel of the trucks in the mine
- **Ammonia production:** domestic demand for production of explosives
- **Production at the mine:** Production at Escondida with the best PV conditions
- **Intermediate location:** Production close to pumping stations.
- **Production at the port:** Close to export port and desalination plant, short distance to CO₂-sources.

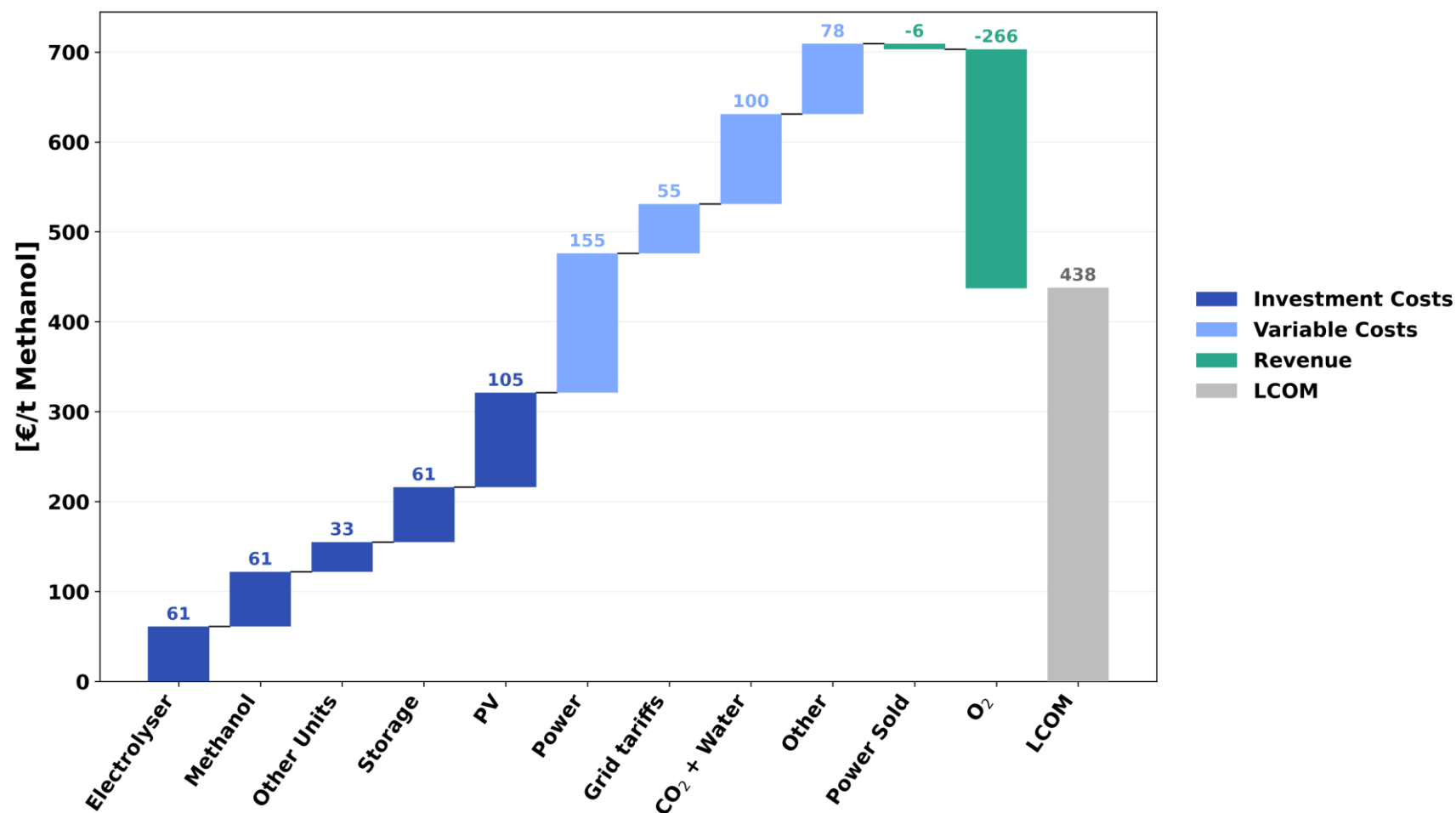
▪ Assumptions

- No use of waste heat possible
- All oxygen can be sold in liquid form

Base Case	Value
Power Price Wholesale Market	Average Price Level (2025) 47.8 Euro/MWh
Grid Tariff	11 Euro/MWh
RES timeseries	% of capacity (2019)
CO ₂ -Supply Price	60.5-66.5 Euro/t (assumption)
Water Price	0.004 Euro/l (assumption)
Oxygen Price	150 Euro/t
Methanol Demand	416,000 t/a
Ammonia Demand	701,000 t/a
Export Share	40 %
PEM Electrolyzer Efficiency	Variable, 0.75 mean
Lifetime	25 years
Interest Rate (WACC)	8%

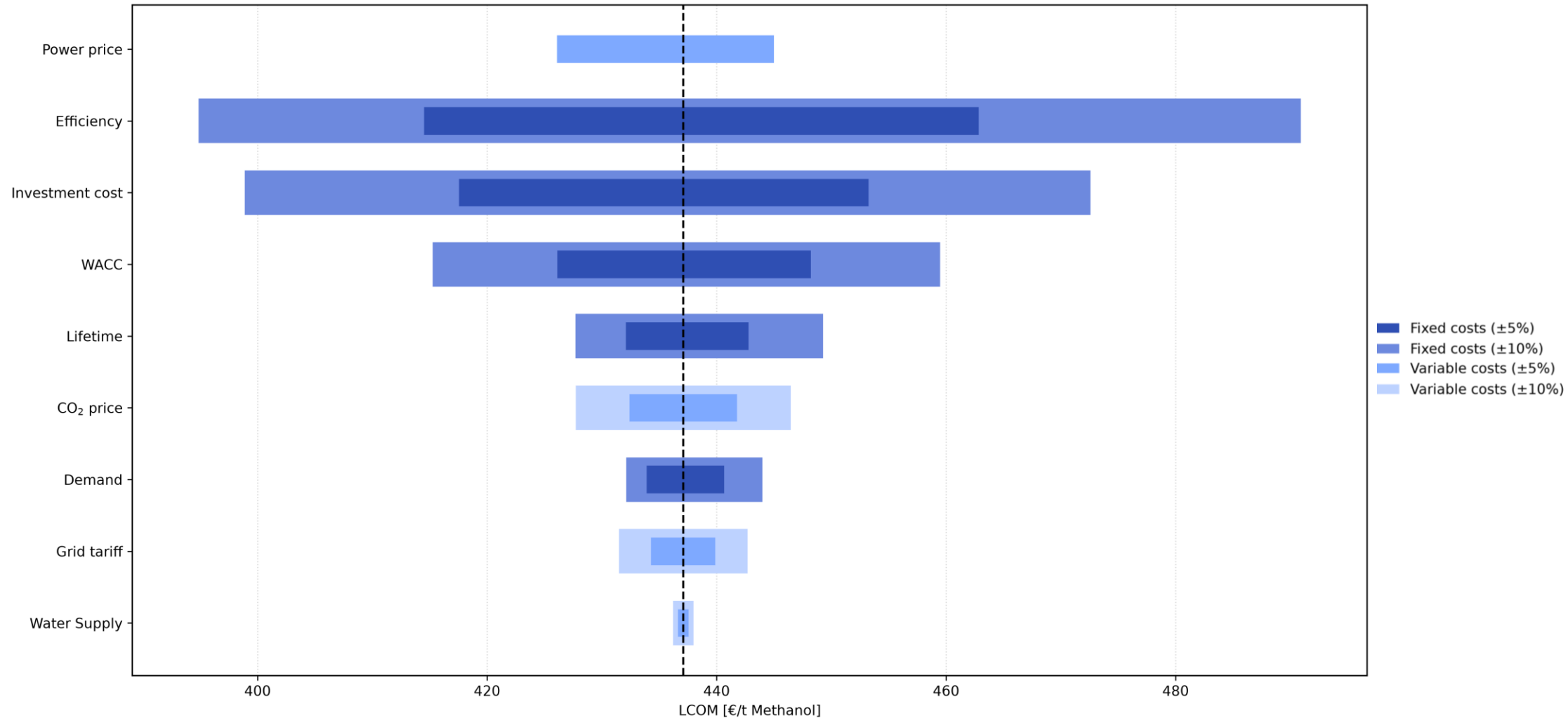
Power Consumption and Oxygen Revenues with highest impact

Revenue from oxygen reduces costs



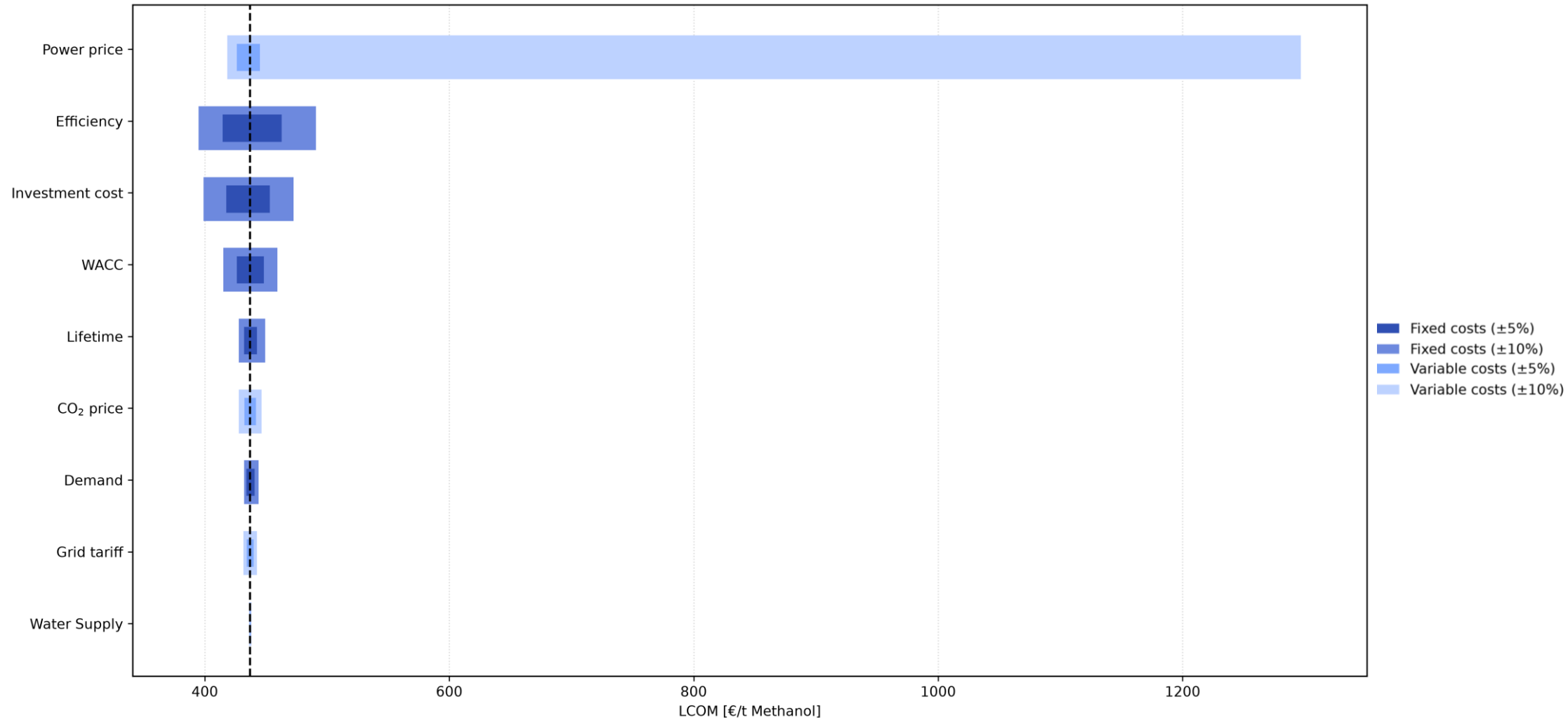
CAPEX seem to drive uncertainty

Sensitivity analysis of the methanol production at Minera Escondida



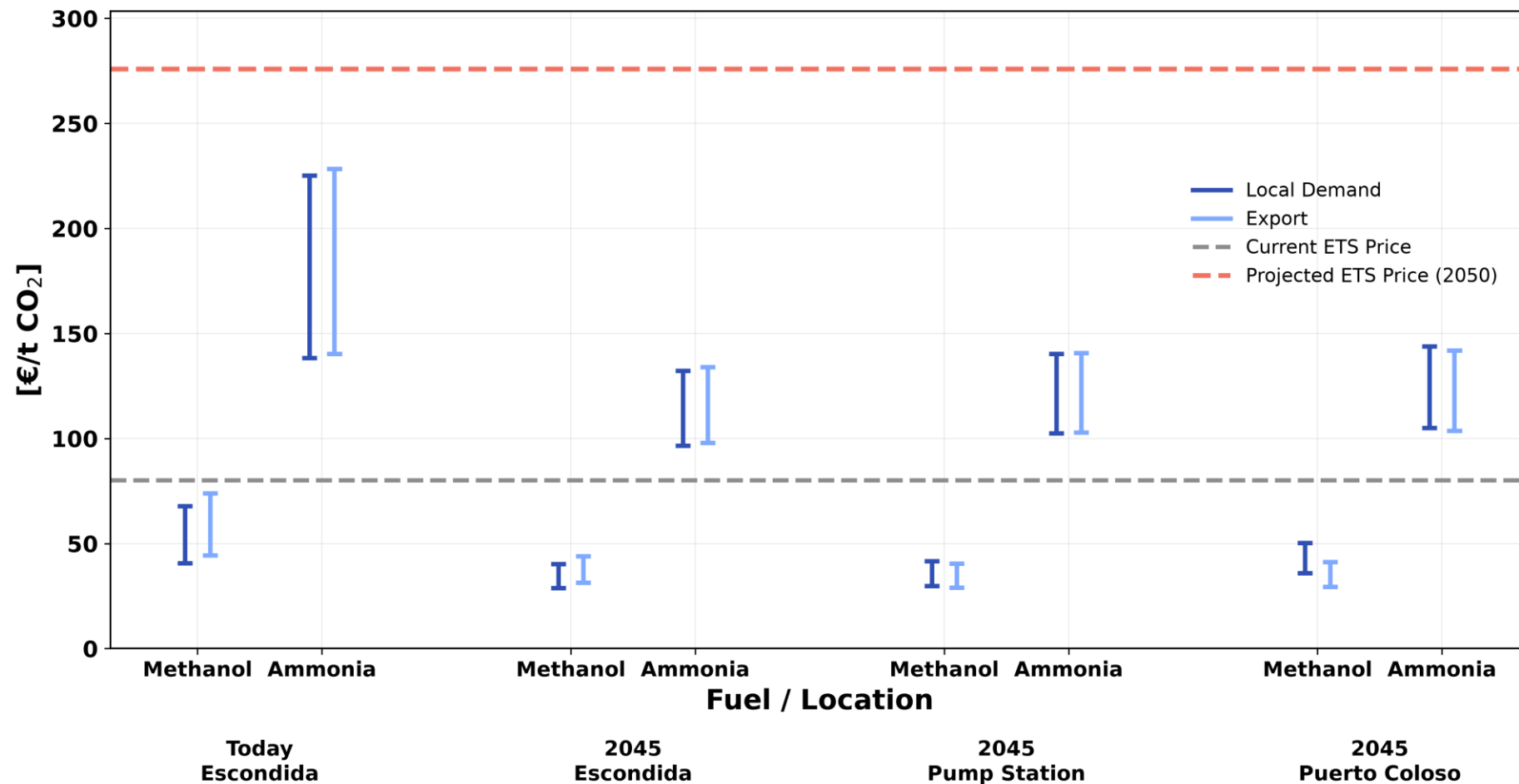
But higher power prices eliminate competitiveness

Sensitivity analysis of the methanol production at Minera Escondida



Carbon pricing is a Key Policy Instrument

Carbon pricing as a measure to accelerate competitiveness



Local demand and carbon pricing are enablers for green fuels

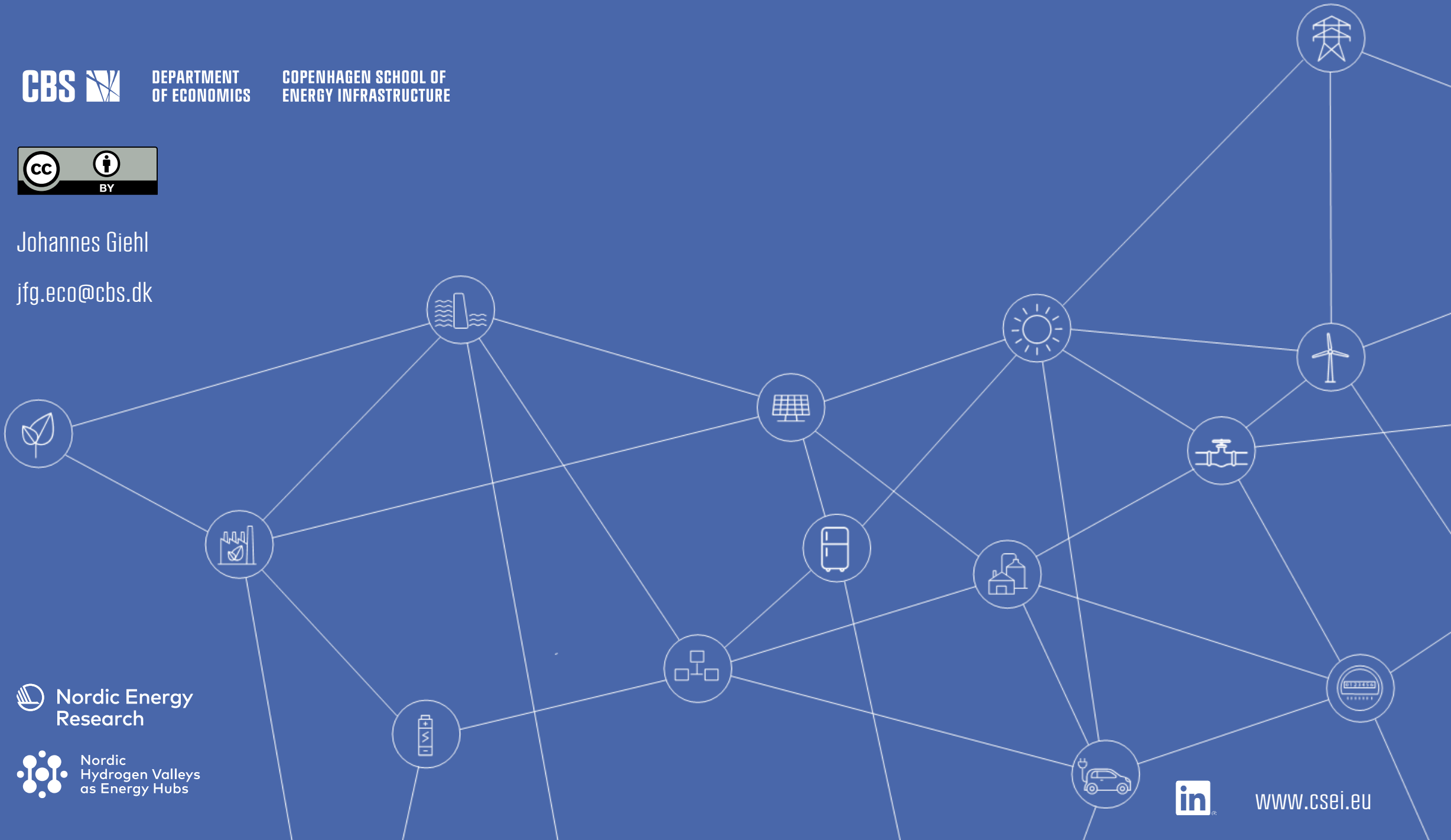
Key Results, Challenges & Limitations

- **Consistent with previous studies**
 - Differences due to modelling approach (e.g., CO₂-import)
- **Mining demand enables early deployment**
- **Carbon pricing substantially improves green fuel competitiveness.**
 - Supports market development
 - Additional demand support needed
- **CAPEX and financing assumptions are dominant cost drivers.**
- **Ammonia benefits from structural independence from CO₂ supply.**
 - Less complex value chain,
 - but significantly higher investment cost
- **Simplified representation of transport infrastructure and logistics.**
 - Simplified CO₂-infrastructure and fuel transportation representation
 - Pipeline and transport investments not modelled
- **Power Price and Market Representation**
 - Need for timeseries of a 100% renewable system
 - Integration of further markets (e.g., system services)
- **100% oxygen utilisation assumed**
 - Internal oxygen demand not considered
 - No analysis of export market potential



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Nordic Energy
Research



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Hydrogen Valleys
as Energy Hubs



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